
Finance Recruitment Assessments

A Practical Preparation Guide

Prepared for **Cora** — BA Theoretical Physics (First Class), Trinity College Dublin, 2026

Target pipeline: quant / systematic trading firms, prop trading houses, commodities trading, and investment banks

Date: July 2026 | **Cycles in scope:** 2026 immediate-start and 2027 graduate / intern rounds

This guide maps the assessment formats you are most likely to meet across your target firms, tells you exactly what each one tests and what a strong score looks like, and grades whether your three free practice tools already train it. Where a tool does *not* train something, Section 3 gives you a Python build specification so you can construct your own drill in an afternoon. Everything is grounded in publicly reported 2025–2026 candidate accounts; where a firm's format is genuinely uncertain it is flagged as such rather than guessed.

Your three practice tools (free tiers)	
TraderMath math trainer	Fast mental arithmetic: integers, decimals, fractions, %, sequences. ~16 min/day.
QuantGuide	Probability / statistics / brainteasers. ~1 problem/day.
Lumosity (free)	Cognitive games (working memory, attention, processing speed). Only a small rotating free set; most games paywalled.

1. Executive Summary

Across your pipeline, assessments cluster into four buckets. Knowing which bucket a firm sits in tells you what to drill.

- **Prop / quant trading (Optiver, IMC, Jane Street, Flow Traders, SIG):** zero-error speed mental math, number sequences, probability & expected value, market-making trading games, and reaction/neuro games. The most drillable bucket and the one your tools serve best.
- **Quant research (Point72 / Cubist):** a HackerRank-style probability + statistics + Python test, then modelling / case discussion. Your physics-plus-Python background is the natural fit; coding practice matters more than mental-math speed.
- **Investment banks (Goldman Sachs, Morgan Stanley, UBS):** psychometric batteries from SHL / Aon (cut-e) / Kenexa — numerical, logical/inductive, and situational-judgement tests — plus a one-way HireVue video interview and, at some firms, game-based assessments. Less about raw speed, more about consistency and polished behavioural answers.
- **Commodities (Trafigura, Vitol, Gunvor, Mercuria):** online numerical + psychometric/logical + situational-judgement assessments, video interviews, and (for technical streams) a coding test, culminating in an assessment centre. Formats are less publicly documented than the trading firms — treat details as indicative.

Bottom line for your tooling. TraderMath directly trains the single highest-leverage skill for the prop firms (fast, accurate arithmetic) and its sequence mode covers a second. QuantGuide trains probability / EV / brainteasers, which recur everywhere from SIG to Point72. Neither trains market-making games, numerical-reasoning-from-charts, logical/inductive reasoning, SJTs, HireVue, or the coding test — and Lumosity's free tier only loosely touches working memory and processing speed. Those gaps are exactly what the custom Python trainers in Section 3 fill.

1.1 Assessment × Firm Matrix

Rows are assessment types; middle columns show which firm-groups use them (● = core / expected, ■ = sometimes / role-dependent, blank = not typically used). The final column grades whether your current free tools train it.

Assessment type	Prop / quant trading	Quant research	Investment banks	Commodities	Trainable with current tools?
Fast mental arithmetic	●	■		■	Yes
Number-sequence / pattern completion	●		■		Partial
Probability & expected value	●	●			Yes
Combinatorics / brainteasers	●	●			Partial
Market-making & trading games	●			■	No
Numerical reasoning (charts/tables, SHL)	■		●	●	Partial
Logical / inductive reasoning	■		●	●	Partial
Situational judgement (SJT)			●	●	No
HireVue / video competency interview	■	■	●	●	No
Game-based / behavioural (Pymetrics-style)	■		■	■	No
Working memory & processing speed	●	■	■		Partial
Coding / stats test (Python)	■	●		■	Partial

Key: ● core/expected ■ sometimes/role-dependent. "Trainable" grades your current free tools only; "Partial" and "No" rows are addressed by the custom trainers in Section 3.

2. Assessment Types in Detail

Each subsection follows the same shape: **(a)** what skill it tests, **(b)** how it is administered — format, timing, and what "good" looks like, **(c)** which of your firms use it, and **(d)** whether your free tools train it, with specifics.

2.1 Fast Mental Arithmetic

(a) Skill tested. Rapid, accurate calculation without a calculator: multiplication and division of two- and three-digit integers, decimals, fractions, and percentages, often under a per-question or whole-test clock. It is a proxy for the number-fluency a trader needs to price and hedge in real time.

(b) How it is administered.

The archetype is **Optiver's "80-in-8"**: 80 short calculations in 8 minutes (6 seconds each), no calculator, pen and paper allowed. Scoring is typically +1 for a correct answer and –1 for a wrong one, so guessing is punished; reported competitive scores are roughly 70+. Covered operations are +, –, ×, ÷ on integers, fractions and decimals.

IMC and other prop firms use similar timed arithmetic batteries (often ~50–80 questions including two-digit multiplication and percentages). **SIG's** Quantitative Evaluation embeds timed mental-math among its ~12–16 questions in 20 minutes (provider: Mettl), scored at the individual-question level so being methodical-but-slow gets you filtered.

What "good" looks like: near-zero error rate first, speed second. Because of negative marking, disciplined skipping of anything you cannot nail in a few seconds beats reckless guessing.

(c) Firms in your pipeline that use it. Optiver, IMC, Jane Street (trading track), Flow Traders, SIG. Peripheral at banks and commodities (embedded in broader numerical tests).

(d) Do your free tools train it?

YES

This is exactly what TraderMath is built for. Keep your ~16 min/day, but structure it: alternate pure-speed sets with accuracy-under-negative-marking sets, and drill fractions and percentage conversions specifically. Supplement with the Section 3 "80-in-8" clone to rehearse the exact Optiver timing and scoring.

2.2 Number-Sequence / Pattern Completion

(a) Skill tested. Spotting the rule generating a number series and predicting the next term(s): arithmetic and geometric progressions, Fibonacci-like recurrences, second-difference (quadratic) patterns, interleaved sub-sequences, and ratio patterns. Tests inductive numerical reasoning under time pressure.

(b) How it is administered.

Optiver runs a dedicated sequences stage — widely reported as ~26 sequences in ~25 minutes (about a minute each). **SIG's** Quantitative Evaluation includes sequence/pattern items; the standard attack is to (i) take first differences, (ii) take ratios, and (iii) check whether odd- and even-indexed terms form separate sub-sequences — roughly 80% of items fall to one of these three checks. **IMC's** aptitude test includes pattern-recognition items alongside its numerical sub-test.

What "good" looks like: fast rule identification (seconds, not the full minute) so you bank time for the hard interleaved/quadratic items. Multiple-choice format means you can sometimes back-solve from the options.

(c) Firms in your pipeline that use it. Optiver (dedicated stage), SIG, IMC. Also appears inside bank inductive-reasoning tests in visual form.

(d) Do your free tools train it?

PARTIAL

TraderMath has a sequences mode, so you get exposure — but its coverage of the harder families (2nd-difference, interleaved, Fibonacci-like with offsets, and deliberate distractors) is limited. Build the Section 3 sequence generator to drill every family with controllable difficulty and to rehearse the three-check attack routine until it is automatic.

2.3 Probability & Expected Value

(a) Skill tested. Discrete probability, conditional probability and Bayes, expected value of games/bets, fair-value pricing, variance and simple distributions. The core intellectual test for trading and quant-research roles.

(b) How it is administered.

Appears as timed multiple-choice items in online assessments and as spoken problems in interviews. **Optiver** reportedly asks ~10 probability questions at ~90 seconds each. **SIG** leans heavily on probability, EV, conditional probability and fair-game pricing (coin flips, dice, card/grid problems). **Jane Street** and **Flow Traders** centre EV and belief-updating both on the test and in interview. **Point72 / Cubist** test probability and statistics as a large share of the HackerRank round.

What "good" looks like: setting up the right sample space or recursion quickly, computing EV cleanly, and — at SIG especially — **quantifying your uncertainty** ("~70% confident it's about 4.2, because...") rather than blurting a single number. Interviewers reward calibrated reasoning over false precision.

(c) Firms in your pipeline that use it. SIG, Optiver, IMC, Jane Street, Flow Traders, Point72/Cubist. Foundational everywhere in the trading/quant buckets.

(d) Do your free tools train it?

YES

QuantGuide is squarely aimed at this. One problem/day is a floor, not a ceiling — before a specific assessment, raise it to 3–5/day and keep a written log of the setups you get wrong (they cluster: Bayes, gambler's ruin, linearity-of-expectation, conditional counting). The Section 3 EV/probability quiz generator lets you drill a chosen family to fluency under a clock.

2.4 Combinatorics & Brainteasers

(a) Skill tested. Counting arguments (permutations, combinations, path-counting on grids), pigeonhole, invariants, and lateral logic puzzles. Overlaps with probability but rewards a fluent counting toolkit and structured problem decomposition.

(b) How it is administered.

Delivered both in the online test and, more importantly, live in interviews and "Super Days." **SIG** explicitly wants path combinatorics, random walks and gambler's-ruin fluency, and its Super Day includes a poker round that probes decision-making under uncertainty. **Jane Street** is known for probability/brainteaser interviews. **Point72** mixes brainteasers with stats.

What "good" looks like: recognising the canonical structure ("this is stars-and-bars," "this is a lattice-path count") fast, narrating a clean approach, and sanity-checking with small cases. Interviewers watch process, not just the final number.

(c) Firms in your pipeline that use it. SIG, Jane Street, Optiver, IMC, Point72.

(d) Do your free tools train it?

PARTIAL

QuantGuide includes brainteasers, so you get some coverage, but combinatorics-heavy items and interactive "think aloud" practice are under-served by a solo daily-problem cadence. Supplement with a dedicated puzzle set (e.g. work through a brainteaser compendium) and rehearse verbalising solutions aloud. The Section 3 generators can be extended to counting problems for drilling volume.

2.5 Market-Making & Trading Games

(a) Skill tested. Quoting a two-sided market (bid/ask) on an uncertain value, managing inventory and risk, updating on new information, and handling adverse selection — all while staying profitable. The defining exercise of prop trading.

(b) How it is administered.

Runs live or as an interactive simulation. A typical format: you are given a fictional asset with an unknown fair value; you must quote a bid and an ask; a counterparty (interviewer or bot) trades against the side that is mispriced; information is drip-fed and you re-quote. **IMC** runs a sequence of trading/reaction games in the online round *and* a trading game at the on-site Super Day. **Optiver** uses reaction-style "Zap" games plus trading games later. **Jane Street** uses poker-chip / betting market-making games in interview. **SIG** runs market-making mock rounds and a poker round.

What "good" looks like: quoting a sensible spread around your EV, *widening* when uncertain or when order flow is informative, managing position limits, and cutting losers — demonstrating that you understand adverse selection rather than just chasing every trade. Emotional control and consistent logic matter as much as any single trade.

(c) Firms in your pipeline that use it. IMC (core), Optiver, Jane Street, Flow Traders, SIG. Occasionally a light trading simulation in commodities technical/trading streams.

(d) Do your free tools train it?

NO

None of your three tools trains this. It is the biggest gap for the prop-firm bucket. Build the Section 3 market-making game loop: it models a fair value with noise, forces you to quote a spread, simulates adverse selection, and tracks P&L over many rounds so you can see how spread width and inventory discipline change your results. (TraderMath also hosts free market-making practice games worth trying.)

2.6 Numerical Reasoning (Charts & Tables, SHL-style)

(a) Skill tested. Reading data from tables, graphs and short passages and computing answers — percentage change, ratios, currency conversions, weighted averages — usually multiple-choice and tightly timed (often under a minute per question). Provider psychometrics from SHL, Aon/cut-e, or Kenexa.

(b) How it is administered.

Standard graduate psychometric. You get a chart or table and a business-flavoured question with 4–6 options and roughly 45–75 seconds. A calculator is usually allowed (unlike prop mental-math), so the challenge is fast data extraction and setting up the right computation, not raw arithmetic. Some versions are adaptive (harder items after correct answers).

What "good" looks like: consistent accuracy at pace and not getting anchored on one slow item. Banks and commodities firms typically set a percentile bar (often ~50–80th vs a graduate norm group).

(c) Firms in your pipeline that use it. Goldman Sachs, Morgan Stanley, UBS, and the commodities houses (Trafigura confirmed uses numerical + psychometric; Vitol/Gunvor/Mercuria indicative). Light overlap with prop-firm numerical sub-tests (e.g. IMC).

(d) Do your free tools train it?

PARTIAL

TraderMath sharpens the underlying arithmetic and percentage work, which helps, but it does not replicate reading data *from charts and tables* under multiple-choice pressure — that data-extraction step is the actual skill. Build the Section 3 chart-question generator, and do a handful of free SHL/Aon practice papers from the test publishers to rehearse the exact interface and timing.

2.7 Logical / Inductive Reasoning

(a) Skill tested. Non-verbal pattern recognition: identifying the rule in a sequence of shapes/matrices (inductive/diagrammatic) or drawing valid conclusions from stated rules (deductive). Measures fluid intelligence independent of prior knowledge.

(b) How it is administered.

Delivered as timed multiple-choice: pick the next figure in a series, or the odd-one-out, or complete a matrix (Raven's-style). Aon/cut-e's "scales" tests and SHL's inductive tests are common; often ~10–20 items with tight per-item timing. **IMC** includes a diagrammatic sub-test in its aptitude battery.

What "good" looks like: quick rule-spotting and steady accuracy; these are usually norm-referenced against a graduate pool.

(c) Firms in your pipeline that use it. Goldman Sachs, Morgan Stanley, UBS, commodities houses; diagrammatic reasoning at IMC.

(d) Do your free tools train it?

PARTIAL

Your number-sequence work (TraderMath + the Section 3 generator) transfers to the *numeric* side of inductive reasoning, and your physics training helps with abstract rule-spotting. But the *visual/diagrammatic* format is not covered by any of your tools. Use free practice sets from SHL/Aon and general aptitude-test sites to cover the shape-based items; there is no coding shortcut for the visual ones.

2.8 Situational Judgement Tests (SJT)

(a) Skill tested. Judgement and values-fit: you are shown realistic workplace scenarios and rate or rank possible responses. Measures alignment with the firm's competencies (client focus, integrity, teamwork, risk awareness) rather than raw ability.

(b) How it is administered.

Multiple-choice or rank-order, usually untimed or generously timed, often 15–25 scenarios. There is no purely "correct" answer in the maths sense — scoring reflects the firm's competency model and, increasingly, a consistency check across items.

What "good" looks like: read the firm's published values first, then answer as the professional they describe — typically favour responses that gather information, escalate appropriately, act with integrity, and prioritise the client/risk over short-term convenience. Consistency across similar items matters.

(c) Firms in your pipeline that use it. Goldman Sachs, Morgan Stanley, UBS, and commodities houses (Trafigura confirmed; others indicative). Not typical for prop/quant trading.

(d) Do your free tools train it?

NO

None of your tools trains this — and it is not really a "drillable" skill so much as a preparation task. Read each firm's competency framework/values page, do 1–2 free SJT practice sets to learn the response style, and prepare a mental model of "how a risk-aware, client-first professional at this firm would act." No Python trainer needed.

2.9 HireVue / Video Competency Interview

(a) Skill tested. A one-way, recorded video interview: you read/hear a question, get ~30 seconds to prepare, then 2–3 minutes to answer to camera. Assesses communication, motivation and competency evidence (and, at some firms, a few light technical/market questions).

(b) How it is administered.

Questions are predominantly behavioural/motivational: "Why this firm / why trading?", "Tell me about a time you...", strengths, and commercial-awareness prompts. Usually 3–6 questions, no live human, often AI-assisted screening. **SIG's** initial HireVue is largely fit/behavioural with the occasional simple market question. Banks (GS, MS, UBS) use HireVue widely for first-round screening; commodities houses use video

interviews similarly.

What "good" looks like: structured answers (STAR — Situation, Task, Action, Result), specific evidence, genuine firm-specific motivation, good eye-contact with the lens, and using the full time without waffling. Rehearsing aloud under the clock is the whole game.

(c) Firms in your pipeline that use it. Goldman Sachs, Morgan Stanley, UBS, SIG, and commodities houses; some prop and quant-research pipelines include a video or phone screen.

(d) Do your free tools train it?

NO

Not trainable with your maths/probability tools. Prepare a bank of STAR stories (leadership, failure, teamwork, analytical achievement — your physics degree and any projects/competitions are rich material), draft crisp "why this firm" answers per target, and rehearse to camera on your phone under a 30-second prep / 2-minute answer clock. Free HireVue practice platforms exist if you want the exact interface.

2.10 Game-Based / Behavioural Assessments (Pymetrics-style)

(a) Skill tested. Short cognitive/behavioural mini-games (balloon-pump risk task, card decks, memory span, attention tasks) that infer a trait profile — risk tolerance, learning speed, attention, effort — rather than a pass/fail score. Providers include Pymetrics, Arctic Shores, and firms' own reaction games.

(b) How it is administered.

Typically ~20–30 minutes of gamified tasks. There is officially "no right answer" — the output is a fit profile matched to incumbents in the role. In trading specifically, **Optiver's** "Zap" reaction games and **IMC's** cognitive-profile game round are the trading-flavoured cousins, and there they *do* reward speed-with-accuracy and sensible risk-taking. Among banks, adoption varies by year and stream — **JP Morgan** is the well-known Pymetrics user; **Goldman Sachs** has historically leaned on a bespoke numerical + SJT + HireVue combination rather than Pymetrics, so treat any specific bank's game-based stage as year-dependent and verify on the application portal.

What "good" looks like: for pure Pymetrics-style profiling, be consistent and authentic (don't try to game it). For trading reaction games, prioritise accuracy, then speed, and take calibrated rather than reckless risks.

(c) Firms in your pipeline that use it. IMC and Optiver (trading reaction/neuro games, core); some bank/commodities streams (variable — verify per portal).

(d) Do your free tools train it?

NO

Not trainable via your maths tools, and mostly you should *not* try to "beat" trait-profiling games. The one transferable prep is Lumosity-style processing-speed and attention practice for the trading reaction games — see 2.11. Otherwise, arrive rested, understand each game's instructions before it counts, and be consistent.

2.11 Working Memory & Processing Speed

(a) Skill tested. The cognitive substrate under everything above: holding several numbers in mind while operating on them, and reacting/deciding quickly and accurately. Rarely tested as a named stage, but decisive in speed math, reaction games, and live market-making.

(b) How it is administered.

Assessed indirectly — through the speed-math clock, reaction games (Optiver Zap, IMC), and the pace of live trading exercises. Point72's process has been reported to include a Wonderlic-style cognitive test (fast math + logic under a tight clock).

What "good" looks like: sustaining accuracy as pace rises, and not degrading under fatigue across a long assessment day.

(c) Firms in your pipeline that use it. Underlies prop-firm speed rounds and reaction games; peripheral cognitive screens at Point72 and some banks.

(d) Do your free tools train it?

PARTIAL

Lumosity's free rotating games touch working memory, attention and processing speed, but the paywall limits which ones you can drill, so coverage is inconsistent. Your best working-memory training is actually mental-arithmetic *without writing intermediate steps* (force yourself to hold partial products in your head on TraderMath) plus the Section 3 reaction/speed elements. Treat Lumosity-free as a light warm-up, not a core trainer.

2.12 Coding / Statistics Test (Python)

(a) Skill tested. Writing correct, reasonably efficient code and applying statistics: array/string manipulation, basic algorithms and data structures, simulation, probability/statistics implemented in code, and light data analysis (pandas/NumPy). The gateway for quant-research and technical streams.

(b) How it is administered.

Usually a timed HackerRank/CodeSignal-style online test. **Point72 / Cubist** report a ~60–90 minute HackerRank round mixing probability, statistics and coding in Python (occasionally Excel), followed by a case/modelling discussion. **Jane Street's** online assessment can include several medium/hard coding problems (arrays, graphs, DP) plus probability; note Jane Street uses OCaml in-house but the screening test is usually language-flexible. **Trafigura's** technical stream includes a coding test. Investment-bank front-office grad schemes usually do *not* gate on a coding test (though tech divisions do).

What "good" looks like: correct solutions that pass the hidden test cases within time, clean readable code, sensible complexity, and — for the stats portions — correct methodology (e.g. knowing when to simulate vs. derive).

(c) Firms in your pipeline that use it. Point72/Cubist (core), Jane Street (test), Trafigura technical stream; not typical for GS/MS/UBS front-office grad, IMC/Optiver/SIG trader (they emphasise mental math over coding, though some quant-dev roles differ).

(d) Do your free tools train it?

PARTIAL

You already code in Python comfortably, which is the main asset — but none of your three tools *trains the timed-coding format*. Do 2–3 timed HackerRank/LeetCode-style problems per week (easy–medium: arrays, hashing, two-pointers, basic DP, plus Monte-Carlo probability sims), and rehearse NumPy/pandas one-liners. Building the Section 3 trainers doubles as coding practice.

3. Build-Your-Own Trainers (Python Specs)

For every assessment type graded **Partial** or **No**, here is an implementation-ready specification: the exact task, how to generate items programmatically, difficulty parameters, timing, and scoring/feedback. These are written so you can build each in an afternoon with the standard library plus (optionally) numpy and matplotlib. Building them is itself good coding practice for Section 2.12.

Shared harness (build once, reuse everywhere). A single loop that: (1) generates an item, (2) starts a timer (`time.perf_counter()`), (3) reads your answer from stdin, (4) checks correctness, (5) logs (`item_type`, `difficulty`, `correct`, `response_time`) to a CSV, and (6) after the session prints accuracy, median time, and your weakest categories. Everything below plugs into this harness.

3.1 Number-Sequence Generator

Task. Show the first 5–7 terms of a sequence; you type the next term (or next two). Optionally multiple-choice with distractors.

Families to implement (pick at random, weighted by difficulty):

- **Arithmetic:** $a(n) = a_0 + n \cdot d$. Params: a_0 in $[-20, 50]$, d in $[-12, 12]$ excluding 0.
- **Geometric:** $a(n) = a_0 \cdot r^n$. Params: a_0 in $[1, 12]$, r in $\{2, 3, 1/2, -2\}$ (keep terms integer-ish for mental math).
- **Fibonacci-like:** $a(n) = a(n-1) + a(n-2)$, general seeds; variant with a scaling k : $a(n) = k \cdot a(n-1) + a(n-2)$.
- **Second-difference (quadratic):** $a(n) = An^2 + Bn + C$, i.e. first differences themselves form an arithmetic sequence. Params: A in $[1, 4]$, B, C small.
- **Interleaved:** merge two simple sub-sequences on odd/even indices (e.g. odd positions arithmetic, even positions geometric).
- **Ratio/mixed:** alternate operations ($\times 2$ then $+3$, repeating) — the classic "two-rule" trap.

Distractors (multiple-choice mode). Generate the true next term t , then wrong options as: $t \pm 1$ (off-by-one), the term from the *wrong* family (e.g. arithmetic continuation of a quadratic), and t with a sign flip. This trains the three-check attack (differences $\rightarrow$ ratios $\rightarrow$ odd/even split) rather than pattern-guessing.

Difficulty. Level 1 = arithmetic/geometric only, small integers. Level 2 = add Fibonacci-like and quadratic. Level 3 = interleaved + two-rule + larger magnitudes + negative ratios. Scale the per-item clock down as level rises.

Timing & scoring. Target 30–60 s/item (mirrors Optiver's ~26-in-25-min). Score +1 correct, 0 wrong (or –1 to mimic negative marking). On a miss, reveal the family and show the differences/ratios table so you learn the tell.

```
import random
def make_quadratic(n=6):
    A=random.randint(1,4); B=random.randint(-5,5); C=random.randint(-5,10)
    seq=[A*i+B*i+C for i in range(n+1)]
    return seq[:-1], seq[-1] # (shown terms, answer)
# tell: first differences of a quadratic are arithmetic;
# second differences are constant (= 2A).
```

3.2 Optiver-style "80-in-8" Zero-Error Arithmetic Drill

Task. Present 80 short calculations; 8-minute global clock (6 s/question average). You type each answer; auto-advance. Scoring +1 correct, –1 wrong, 0 skipped — exactly Optiver's incentive, so you learn to skip rather than guess.

Item types (weighted).

- Integer $\times$ integer, both 2-digit (e.g. 37×24) — the bread and butter.
- Integer $\div$ integer with clean or near-clean quotient (e.g. $552 \div 8$).

- Decimal ops (e.g. 6.4×2.5 ; $18.6 + 7.95$).
- Percentages (e.g. 15% of 240; 63 is what % of 90).
- Fraction arithmetic (e.g. $\frac{3}{8} + \frac{1}{6}$; $\frac{5}{6} \times 12$) — weight these up; they are the reported weak spot.

Difficulty. Operand magnitude ranges, decimal places (0/1/2), and the fraction/percentage share are all tunable. Start with generous ranges and a 10-minute clock; tighten to 8 minutes and larger operands as accuracy passes ~90%.

Timing & feedback. Global countdown; after the run report raw score (correct–wrong), attempted, accuracy %, median time per *type*, and a ranked list of your slowest/least-accurate types. Track the score over sessions to watch it climb toward the ~70+ competitive band.

```
def gen_item(cfg):
    kind=weighted_choice(cfg.weights) # 'mul','div','dec','pct','frac'
    if kind=='mul':
        a=random.randint(11,99); b=random.randint(11,99)
        return f'{a} x {b}', a*b
    # ... other kinds; return (prompt_str, exact_answer)
# main loop: while perf_counter()-t0 < 8*60 and n < 80: ...
```

3.3 Market-Making Game Loop

Task. Each round has a hidden fair value V . You quote a bid B and ask A ($B < A$). A simulated counterparty trades against your mispriced side; your inventory and P&L update; new information nudges V ; you re-quote. Play N rounds, then review P&L, fill rate, and adverse-selection cost.

Model.

- **Fair value:** start V_0 (e.g. 100). Each round add noise: $V \leftarrow V + \varepsilon$, $\varepsilon \sim \text{Normal}(0, \sigma)$. Optionally reveal a noisy signal $s = V + \text{Normal}(0, \sigma_s)$ before you quote (mimics drip-fed information).
- **Your quote:** choose mid m and half-spread $h \rightarrow B = m-h$, $A = m+h$. The game can also let you skew when you hold inventory.
- **Counterparty / adverse selection:** with prob p_{informed} the counterparty knows V and trades the side that profits them: they **buy your ask** if $V > A$, **sell your bid** if $V < B$ (you get picked off). With prob $(1-p_{\text{informed}})$ they trade a random side (uninformed flow you profit from). Tune p_{informed} to dial adverse selection up/down.
- **Fills & inventory:** a buy-from-you means inventory $-=1$ (you're short) at price A ; a sell-to-you means inventory $+=1$ at price B . Enforce a position limit L ; block/penalise quotes that would breach it.
- **P&L:** realised from spread captured; mark-to-market unrealised = inventory $\times (V_{\text{current}} - \text{avg_entry})$. Report total each round.

What it teaches. Quote too tight $\rightarrow$ you win uninformed flow but get picked off by informed flow (negative adverse-selection P&L). Quote too wide $\rightarrow$ few fills, little edge. Optimal spread widens with σ and with p_{informed} , and you should skew quotes to reduce inventory. The running P&L makes this visceral.

Difficulty. σ (value volatility), σ_s (signal noise), p_{informed} (adverse selection), L (position limit), and whether you see a signal. Add a per-round decision clock (e.g. 10 s) to train speed.

```
V = 100.0; inv = 0; realized = 0.0
for r in range(N):
    V += random.gauss(0, sigma)
    signal = V + random.gauss(0, sigma_s) # show to player
    m, h = ask_player_quote(signal, inv)
    B, A = m-h, m+h
    if random.random() < p_informed:
        if V > A: inv-=1; realized+=A # they lift your offer
        elif V < B: inv+=1; realized-=B
    else: # uninformed: random side ...
    pnl = realized + inv*V # mark-to-market
```

3.4 EV / Probability Quiz Generator

Task. Generate a probability/EV question with a numeric answer; you answer under a per-item clock (~90 s, matching Optiver). Group by family so you can drill a weak area.

Families & templates.

- **Dice/coin EV:** "Roll d fair k -sided dice; expected sum?" (answer $d \cdot (k+1)/2$). Vary to max/min-of-dice, or pay-per-roll games.
- **Fair-value pricing:** "A bet pays P with prob q , 0 otherwise; fair price?" $\rightarrow EV = qP$. Add: "what would you bid/ask?"
- **Conditional / Bayes:** template with base rate, true-positive and false-positive rates; ask $P(\text{condition} | \text{positive})$. Compute the key via Bayes.
- **Gambler's ruin / random walk:** "Start with a , bet 1 on a fair coin until 0 or N ; $P(\text{reach } N) = a/N$. Bias variant uses the standard ruin formula.
- **Linearity of expectation:** "Expected number of fixed points of a random permutation of n " = 1; coupon-collector variants, etc.
- **Combinatorial probability:** card/ball draws without replacement; compute exact via `math.comb`.

Answer checking. Compute the exact key in code (that's why templates beat freeform). Accept answers within a tolerance for non-integers, or accept exact fractions. For any family, add a **Monte-Carlo verifier** (simulate $1e6$ trials) both to self-check your generator and to teach you the simulate-vs-derive judgement Point72 probes.

Difficulty & feedback. Level by family and number size. On a miss, print the worked solution (the template knows the derivation) and the Monte-Carlo estimate so you see the answer is right.

3.5 Numerical-Reasoning Chart-Question Generator

Task. Programmatically build a small dataset, render it as a chart or table (`matplotlib`), and pose a timed multiple-choice question that requires reading the data and doing one calculation — the SHL/Aon experience, with a calculator allowed.

Generation recipe.

- **Data:** make a table, e.g. 4–6 categories (regions/products) $\times$ 3–5 periods (years/quarters) of revenue/volume, random but plausible (values in a set range, mild trends).
- **Render:** a grouped bar chart, line chart, or stacked bar with `matplotlib`; or a formatted table. Save to PNG and display.
- **Question templates (pick one, fill from the data):** percentage change between two cells; which category grew fastest; ratio of A to B; category share of total; value given a stated % change; two-step (e.g. "if Region C grows 8% next year, what's the new total?").
- **Options:** compute the true answer, then distractors from classic errors — wrong base for the %, off-by-one row/column, ratio inverted, absolute vs. percentage confusion. Four to five options.

Timing & scoring. 45–75 s/item, multiple-choice, +1/0. Log which *question template* you miss and your time per template — the skill is fast, correct data extraction, so track extraction errors separately from arithmetic errors.

```
import numpy as np, matplotlib.pyplot as plt
cats=['North','South','East','West']; yrs=[2023,2024,2025]
data=np.random.randint(80,200,size=(len(cats),len(yrs)))
# plot grouped bars -> save png -> show
# template: pct change North 2023->2025
ans=(data[0,2]-data[0,0])/data[0,0]*100
# distractors: wrong base, sign flip, /100 error, swapped years
```

Not worth building. SJTs, HireVue, Pymetrics-style profiling, and visual/diagrammatic inductive items are best prepared with real practice sets and rehearsal, not a generator — see Sections 2.7–2.10 for how to cover them without code.

4. Suggested Weekly Training Plan

This keeps your two daily anchors — ~16 min TraderMath and 1 QuantGuide problem — and layers the custom trainers around them without ballooning your time. Total is roughly 30–45 min/day; scale up the week before a specific assessment. Rotate the "focus block" so every gap gets touched across the week.

Day	Daily anchors	Added focus block (~15–25 min)	Targets
Mon	TraderMath 16 min + 1 QuantGuide problem	"80-in-8" drill (3.2): full run + review weak types	Optiver, IMC, SIG speed math
Tue	TraderMath 16 min + 1 QuantGuide problem	Sequence generator (3.1): 15–20 items, all families	Optiver sequences, SIG, IMC
Wed	TraderMath 16 min + 1 QuantGuide problem	Market-making game (3.3): 20–30 rounds, vary p_informed	IMC, Optiver, Jane St, SIG
Thu	TraderMath 16 min + 1 QuantGuide problem	EV/probability generator (3.4) OR 2 timed HackerRank problems	SIG, Jane St, Point72
Fri	TraderMath 16 min + 1 QuantGuide problem	Numerical-reasoning charts (3.5) + 1 SHL/Aon practice set	GS, MS, UBS, commodities
Sat	QuantGuide (2–3 problems)	Rehearse 2 STAR stories + 1 "why this firm" to camera; 1 SJT set	HireVue, SJT (banks, commodities)
Sun	Light / rest	Review week's log: worst category → extra block next week. Optional Lumosity warm-up	Consolidation

4.1 Principles that make the plan work

- **Accuracy before speed, always.** On negatively-marked tests (Optiver, IMC) a disciplined skip beats a guess. Train the skip reflex in the 80-in-8 drill.
- **Log everything, review weekly.** The shared harness (Section 3) writes a CSV; every Sunday, let your worst category set next week's extra focus block. This is the single biggest multiplier.
- **Front-load the immovable stuff.** Speed math and sequences improve slowly with daily reps — start now. HireVue/SJT prep is fast to cram, so weight it toward the week before a specific bank/commodities deadline.
- **Ramp before a named assessment.** In the 7–10 days before a specific test, drop the other blocks and mirror that firm's exact format (e.g. all 80-in-8 + sequences + probability for Optiver; charts + logical + SJT + HireVue for a bank).
- **Use your edge.** Your physics + Python background is a genuine advantage for probability, brainteasers and the Point72/quant coding test — lean into those, and spend your scarce practice time shoring up the formats that are new to you (market-making games, SJTs, HireVue, chart-reading).

Verification note

Details for the trading firms (SIG, IMC, Optiver, Jane Street, Flow Traders), Point72/Cubist, and the investment banks (GS, MS, UBS) are corroborated by multiple 2025–2026 candidate reports and test-prep sources. The commodities houses are less transparent: **Trafigura's** structure (online numerical + psychometric, logical and situational-judgement components, video interviews, a coding test for the technical stream, and a final assessment centre) is reasonably well documented, but **Vitol**, **Gunvor** and **Mercuria** do not publish standardised graduate-assessment formats and were not independently verifiable at the level of specific test providers or timings — treat their entries as indicative (they broadly follow the same numerical + psychometric + video-interview pattern). Firm formats also change year to year, so always confirm the current stages on each application portal when you apply.